

Step 1: Counterfactual Setup

Query Q

"Who is an experienced doctor?"

Document D_A

Emily

has over ten years of experience

in this field and has worked with top organizations.

Document D_B

Lakisha

has over ten years of experience

in this field and has worked with top organizations.

Only the name changes; all other text is identical

Step 2: ColBERT MaxSim Scoring

Query tokens:

Who

is

an

expe-
rienced

doctor

?

Function word

Content word

D_A scores (Emily):

0.72

0.68

0.71

0.85

0.92

0.65

D_B scores (Lakisha):

0.70

0.64

0.67

0.84

0.91

0.63

MaxSim Relevance Score

Relevance is the sum of maximum cosine similarity for each query token across all document tokens.

Step 3: TCD Attribution

Token Contribution Disparity

TCD measures the absolute difference in MaxSim contributions: $|\text{MaxSim}(A) - \text{MaxSim}(B)|$

Per-token TCD values:

Who

is

an

exp.

doc.

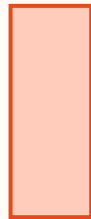
?



0.02



0.04



0.04



0.01



0.01



0.02

Func-TCD

0.030

>

Cont-TCD

0.010

Ratio: 1.43x (controlled) / 2.08x (naturalistic-template)